FDS EXPERIMENT-13

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Experiment to handle missing values and inappropriate data in a given data set.

```
import pandas as pd
import numpy as np
```

```
data = {
    "Name": ["Arun", "Bala", "Charan", "Divya",
             "Ezhil", "Bala"],
    "Age": [20, 21, np.nan, 22, -5, 21],
    "Marks": [85, 90, 75, np.nan, 120, 90],
    "Department": ["CSE", "cse", "ECE",
                   "ECE", "CSE", "cse"]
}

df = pd.DataFrame(data)

print(df)
```

```
     Name   Age  Marks Department
0    Arun  20.0   85.0        CSE
1    Bala  21.0   90.0        cse
2  Charan   NaN   75.0        ECE
3   Divya  22.0    NaN        ECE
4   Ezhil  -5.0  120.0        CSE
5    Bala  21.0   90.0        cse
```

```
df["Age"] = df["Age"].fillna(df["Age"].median())

df["Marks"] = df["Marks"].fillna(
    df["Marks"].median()
)

print(df)
```

```
      Name   Age  Marks Department
0     Arun  20.0   85.0        CSE
1     Bala  21.0   90.0        cse
2   Charan  21.0   75.0        ECE
3    Divya  22.0   90.0        ECE
4    Ezhil  -5.0  120.0        CSE
5     Bala  21.0   90.0        cse
```

```
df.loc[df["Age"] < 0, "Age"] = np.nan

df.loc[
    (df["Marks"] < 0) | (df["Marks"] > 100),
    "Marks"
] = np.nan


df["Age"] = df["Age"].fillna(df["Age"].median())
df["Marks"] = df["Marks"].fillna(df["Marks"].median())

print(df)
```

```
      Name   Age  Marks Department
0     Arun  20.0   85.0        CSE
1     Bala  21.0   90.0        cse
2   Charan  21.0   75.0        ECE
3    Divya  22.0   90.0        ECE
4    Ezhil  21.0   90.0        CSE
5     Bala  21.0   90.0        cse
```

```
df["Department"] = df["Department"].str.upper()

print(df["Department"])
```

```
0    CSE
1    CSE
2    ECE
3    ECE
4    CSE
5    CSE
Name: Department, dtype: object
```

```
df = df.drop_duplicates()
```

```
print(df)
print(df.isnull().sum())
print(df.duplicated().sum())
```

```
     Name   Age  Marks Department
0    Arun  20.0   85.0        CSE
1    Bala  21.0   90.0        CSE
2  Charan  21.0   75.0        ECE
3   Divya  22.0   90.0        ECE
4   Ezhil  21.0   90.0        CSE
Name          0
Age           0
Marks         0
Department    0
dtype: int64
0
```